Reg. No. | | | | | | | | | | | [17CE42]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

SOIL MECHANICS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions, choosing at-least **TWO** from each part.
2. Graph sheet can be provided on request
3. Missing data, if any may suitably be assumed.

PART-A

1. a. Explain Atterberg limits. **06**

   b. Define the following: **08**

   i. Voids ratio

   ii. Porosity

   iii. Degree of Saturation

   iv. Percentage of air voids.

2. a. Define the following : **06**

   i. Shrinkage ratio

   ii. Volumetric shrinkage

   iii. Linear shrinkage

   B The plastic limit of a soil is 25% and its plasticity index is 8%. When the soil is dried from its state at plastic limit, the volume change is 25% of its volume at plastic limit (W_p). Similarly the corresponding volume change from the liquid limit to the dry state is 34% of its volume at liquid state (W_L). Determine the shrinkage limit and the shrinkage ratio. **08**

3. a. With a neat sketch explain Darcy's law. **06**

   b. In a falling head permeability test, the initial head (t=0) is 40 cm. the head drops by 5 cm in 10 minutes. Calculate the final head to be at 20 cm. If the sample is 6 cm is height and 50 cm^3 in cross sectional area, calculate the coefficient of permeability, taking cross sectional area of stand pipe is = 0.5 m^2 **08**

4. a. Explain the factors affecting compaction of soil. **08**

   b. A cohesive soil yields a maximum dry density of 1.8 g/cc at an OMC of 16% during a standard proctor test. If the values of G are 2.65, what is the degree of saturation? What is the maximum dry density it can further compacted to? **06**

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PART-B

5. a. Derive an expression to find the effective pressure for two conditions of soil water systems given; **06**

   i. Submerged soil mass

   ii. Saturated soil capillary fringe.

   b. The water table in a certain area is at a depth of 4 m below the ground surface. To a depth of 12 m, the soil consists of very fine sand having an average voids ratio of 0.7. Above the water table the sand has an average degree of saturation of 50%. Calculate the effective pressure on a horizontal plane at a depth 10 m below the ground surface. What will be the increase in the effective pressure if soil gets saturated by capillarity up to a height of 1 m above the water table? Assume G= 2.65 **08**

6. a. Explain Terzaghi's theory of one dimensional consolidation. **08**

   b. A clay layer, whose total settlement under a given loading is expected to be 12 cm settles 3 cm at the end of one month after the application of load increment. How many months will be required to reach a settlement of 6 cm? How much settlement will occur in 10 months? Assume the layer to have double drainage. **06**

7. a. Explain Mohr-Coulomb Failure theory. **07**

   b. Explain triaxial compression test method. **07**

8. a. Explain the stability analysis of finite slopes. **08**

   b. A new canal is excavated to a depth of 5 m below ground level through a soil having the following characteristics. c=14 k N/m^2; ϕ= 15°; e= 0.8 and G=2.70. The slope of banks is 1 in 1. Calculate the factor of safety with respect to cohesion when the canal runs full. If it is suddenly and completely emptied, what will be the factor of safety. **06**